

Abhishek Singh

Data Engineer | Research Informatics | Scientific & Drug Discovery Data Platforms

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PROFESSIONAL SUMMARY

Data Engineer with **4.5+ years of experience** building scientific data platforms and research pipelines for pharmaceutical R&D.; Currently in **Research Informatics at Eli Lilly R&D**;, designing scalable data ingestion pipelines and data lakehouse architectures that integrate experimental and laboratory datasets for drug discovery and early-stage research teams. Strong enterprise background across Snowflake, GCP, and AWS with hands-on experience in DataOps, CI/CD, and cross-functional R&D; team collaboration. Passionate about building data infrastructure that accelerates scientific discovery and data-driven drug development.

PROFESSIONAL EXPERIENCE

Eli Lilly and Company

Apr 2024 – Present

Data Engineer – Research Informatics

Bengaluru, India

- Build and maintain production **Apache Airflow DAGs** for end-to-end ingestion of scientific, experimental, and genomics datasets supporting drug discovery workflows across multiple R&D; teams.
- Design **data lakehouse architecture** on AWS (S3, Glue, Athena) using **Apache Iceberg**, enabling time-travel queries and schema evolution for large-scale research datasets.
- Develop and manage **dbt transformation models on Trino**, integrating experimental and laboratory datasets consumed by large cross-functional R&D; teams.
- Collaborate with bioinformaticians and scientists to build scalable ELT pipelines for genomics and lab data, enabling efficient exploration and analysis of research data.
- Drive DataOps standards: CI/CD for pipeline deployment, dbt data quality tests, data lineage documentation, and monitoring across the research data platform.
- Develop and manage data platforms and applications integrating experimental datasets used by large cross-functional R&D; teams globally.

DataGrokr Analytics

Dec 2021 – Feb 2024 (2 yrs 3 mos)

Data Engineer – Enterprise Data

Bengaluru, India

- Migrated 3 years of historical data for the **largest US grocery chain** from CSV/JSON formats to Snowflake using SnowSQL and Pandas; fully automated the workflow using GitHub Actions.
- Designed and maintained daily sales ELT pipeline in Snowflake using Snowpipe, external stages, streams, tasks, and materialized views; integrated multi-source data for analytics and Power BI reporting.
- Delivered **~\$100K OpEx savings** through cross-country data analysis and cost optimization in Snowflake, in collaboration with cross-functional international teams.
- Built and managed enterprise data pipelines across Snowflake, BigQuery, and GCP (Dataflow, GCS, Cloud Composer/Airflow, Kafka); integrated Cloud Functions with ServiceNow API for pipeline incident management.
- Automated **Snowflake-to-BigQuery migration** pipeline using GitHub Actions, gcloud SDK, Cloud Composer, and Dataflow.
- Developed live **Power BI dashboards** for data lifecycle monitoring (latency, throughput) and spearheaded DataOps and CI/CD standards across the engineering team.
- Managed diverse Snowflake data pipelines ensuring SCD2 table integrity, operational data quality, and financial reporting accuracy.

DataGrokr Analytics

Jun 2021 – Dec 2021 (7 mos)

Data Engineer Intern

Bengaluru, India

- Built automated end-to-end ETL data pipeline using Snowflake, AWS Lambda, and S3 as part of a structured data engineering internship.

TECHNICAL SKILLS

Orchestration & Streaming	Apache Airflow · Apache Kafka · Google Cloud Composer
Data Transformation	dbt (Data Build Tool) · SQL · Pandas · ELT / ETL
Data Platforms & Query	Apache Iceberg · Lakehouse · Trino · Snowflake · BigQuery · ClickHouse · MySQL · PostgreSQL
Cloud – AWS	S3 · Glue · Athena · Lambda · EC2
Cloud – GCP	BigQuery · Dataflow · GCS · Cloud Composer · Cloud Functions
Programming & Scripting	Python · SQL · Shell / Bash Scripting · Linux
Domain Expertise	Drug Discovery Data Platforms · Research Informatics · Scientific Data Platforms · Genomics Pipelines · Biomedical Data Modeling
DevOps / DataOps	GitHub Actions · Git · Docker · CI/CD · Monitoring · Query Optimization · High Performance Computing (HPC)
Visualization & Reporting	Microsoft Power BI · Dashboard Design · Data Analysis
Tools & Productivity	Snowpipe · SnowSQL · Data Modeling · ServiceNow · Jira · Confluence · Automation

PROJECTS

LLM Data Harmonizer

Automated schema detection, vector embedding generation, and schema matching using Large Language Models. Outputs production-ready dbt SQL models — directly applicable to clinical and research data harmonization in pharma settings.

Technologies: Python · LLM · dbt · Vector Embeddings | github.com/AbhishekSingh1180/LLM-data-harmonizer

JiraPilot – LLM-Powered Code Automation

Automated workflow that converts Jira tickets into boilerplate code using an LLM, triggered by Jira Automation and seeded with real repository context via the GitHub API for accurate, project-aware code generation.

Technologies: Python · LLM · Jira API · GitHub API | github.com/AbhishekSingh1180/JiraPilot

Snowflake to BigQuery Migration Pipeline

GitHub Actions-automated GCP data pipeline for Snowflake-to-BigQuery migration using Cloud Composer (Airflow), Dataflow, and GCS with full workflow orchestration and data transformation.

Technologies: GCP · Cloud Composer · Dataflow · Snowflake · GitHub Actions | github.com/AbhishekSingh1180/snowflake-bigquery-data-migration

ELT Pipeline – Snowflake and Airflow

End-to-end ELT data pipeline integrating Snowflake and Apache Airflow for automated data ingestion, transformation, and loading with scheduled orchestration.

Technologies: Snowflake · Apache Airflow · Python · SQL

CERTIFICATIONS

GCP Associate Cloud Engineer	Google Cloud	Sep 2023 – Sep 2026 · Active · google.accreditable.com/0f00628e-2f07-4c5f-b3f5-e6525971ed5c
Snowflake SnowPro Core Certification	Snowflake	Jun 2023 – Jun 2025 · achieve.snowflake.com/704d646d-089a-482a-bcfe-8036db660f30
SQL (Intermediate) Certificate	HackerRank	

EDUCATION

B.Tech (Hons.), Computer Science and Engineering

2018 – 2022

Lovely Professional University · 80.6%

LANGUAGES

English — Professional Proficiency · **Hindi** — Native / Full Proficiency